

September 2, 2026

Dear Hiring Team,

I am writing to apply for the Data Engineer position at Samsara. With a Master of Science in Computer Science from the University of Calgary and hands-on experience building Python, C++, SQL, and data-processing systems for large scientific datasets, I am excited by the opportunity to help Samsara build reliable data pipelines for analytics, model training, and product decision-making.

Samsara's mission immediately caught my attention because it connects software and data to physical operations that matter: transportation, field services, construction, manufacturing, safety, efficiency, and sustainability. I am especially drawn to the idea of working with data from IoT devices and software products, where the quality of computed tables and pipelines directly affects dashboards, analysis, machine-learning workflows, and customer outcomes. That kind of practical, high-impact data engineering is exactly the direction I want to grow in.

My graduate research focused on scalable management of Earth observation and geospatial data. I built C++ and Python software for ingesting, transforming, storing, and retrieving large satellite-imagery datasets, including automated preprocessing workflows with GDAL, Rasterio, SNAP, and satellite data APIs. I also designed a DGGS-based data framework that converted imagery into structured tensor datasets, reduced storage requirements by 38% on a Canada-wide dataset, and improved processing performance through multithreaded implementation. This work gave me practical experience with high-volume data transformation, storage design, validation, time-series retrieval, and the tradeoffs involved in making complex data reliable and usable at scale.

I also bring strong Python, SQL, and backend experience. As a back-end developer, I built Django REST APIs backed by PostgreSQL and implemented production-facing logic for authentication, service plans, and payment records. As a teaching assistant for Developing Big Data Applications, Working with Data at Scale, and Database Management Systems, I mentored students on cloud storage, AWS-based data pipelines, scalable workflows, SQL, and debugging. These experiences align well with Samsara's need for someone who can work across data engineering, software fundamentals, version control, APIs, documentation, and collaboration with Data Science, Analytics, AI/ML, and product teams.

I have not yet owned production Spark orchestration at Samsara's scale, but I have built the foundation that would let me ramp quickly: Python, SQL, data modelling, large-scale data processing, Git-based engineering, cloud-data coursework and mentoring, and a research background that required learning new systems deeply and applying them carefully. I would bring a strong growth mindset, attention to data quality, and real enthusiasm for building long-term data infrastructure that supports both internal teams and customer-facing outcomes.

Thank you for considering my application. I would welcome the opportunity to discuss how my data engineering, software development, and applied research background can contribute to Samsara's Data team.

Sincerely,

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